

Institute of Actuaries of India

Subject SA2 – Life Insurance

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i)
- CFO to be a member of the WPC compulsorily
 - Eligibility criteria prescribed for the Independent Actuary on the WPC
 - Life insurers can add more experts in the WPC – eligibility criteria applicable for the independent actuary are not mandatory for other additional members.
 - Minimum requirements for the contents in the WPC report mandated
 - Appropriateness of the Methodology and basis used in calculation of asset shares, and justification for any change.
 - Bonus earning capacity including its calculation.
 - Sensitivity analysis of bonus rates and basis as appropriate.
 - A brief note on how Policyholders' reasonable expectations (PRE) is met.
 - Any change in special surrender value with justification.
 - Treatment of Fund for Future Appropriation.
 - The expenses debited to the With Profit fund and its appropriateness
 - Report of the WPC to be shared with the Board
 - Clarification that the AA retains the final say whether to change his/her working

[3]

- ii)
- Non-Participating
 - In non-par savings products, both premiums and benefits (death, surrender and maturity) are guaranteed at the outset. There are no discretionary benefits.
 - 100% of the surplus belongs to Shareholders.
 - 100% investment surplus belongs to Shareholders.
 - Surrender values are generally lower than asset shares and 100% of the lapse/surrender gains belong to Shareholders.
 - 100% of the mortality surplus belongs to Shareholders. Mortality improvement is generally not considered in the pricing and may lead to additional unexpected surplus.
 - 100% of the expense gains go to Shareholders' account.
 - Companies might be pricing the products at higher margin due to higher risk and in-built guarantees.
 - Derivatives like FRA may help limit interest-rate risk and may lead to higher MCVNB.
 - Participating
 - 90:10 gate limits the profitability of participating products,
 - Investment surplus is shared in that proportion
 - Surrender profits belong to the fund and shared in the same ratio
 - Similarly for mortality and expense surpluses
 - However, 100% of the deficit is to be made good by shareholders and repatriation from future surplus is not permitted

[0.5 mark for each valid point; Max 3]

[3]

iii)

a)

Risk that interest rate may go down in future and money invested in future does not yield sufficient returns. Ways to manage interest rate risk:

- Better Asset Liability Matching
 - Duration matching [0.5]
 - Assets may have lower duration than liabilities
 - Asset duration can be increased by investing in long duration assets or zero-coupon bonds
 - Cashflow matching [0.5]
- By investing in partially paid bonds or institutional recurring deposits that offer guaranteed known returns on agreed amounts of investment in the future for a defined period [0.5]
- Structuring products to have lower sensitivity to interest rate changes [0.5]
 - Offering single pay versions to minimize reinvestment risk
 - Offering low premium-payment terms
 - Offering short duration covers
 - Offering regular incomes matching with coupon payments and also to reduce the liability duration; etc [0.5]
- Financial derivatives which allow companies to invest new money in the future at pre-agreed investment return. [0.5]

[3]

b)

Financial derivatives

- IRDAI permits FRAs, IRS and IRFs [0.5]
- Hedging allowed for in-force business [1]
 - Reinvestment risk on maturity
 - Reinvestment risk on coupons or interest-income of existing assets
 - Investment of new money when renewal premiums are received
- Hedging not allowed for future new business [0.5]
- Use of derivatives must be only for hedging and not for yield enhancement [0.5]
- If interest rate goes down, the counterparty makes up for the loss [0.5]
- Governance [0.5]
 - IRDAI requires board-approved risk management policy
- To note, downside risk is mitigated but upside potential is also limited [0.5]

[4]

Residual risk

- Hedging is approximate
- Assumptions for future lapse/surrender may not realize
- For new business, there may be a delay in finding suitable derivatives

[Max 1]

[1]

New risks

- Counterparty risks: risk of default

- Governance/Legal/System risks
 - Persistency risk: mis-estimation risk
 - Liquidity risk: not able to buy or sell assets on right time due to volatile market
 - Mark to market impact on solvency
 - Risk of having to unwind hedging
- [1 mark for any four risks]*

[6]

iv)

- a) Limited experience
 - Company does not have any non-linked index-linked product already.
 - The experience on similar products may not be true for the index-linked product.
- b) Product structure and profitability
 - Profitability from all sources to be ensured: mortality, lapse/surrender, investment, expenses.
 - How the benefits are linked to external index can have an impact on profitability.
 - Due to new product category, company may consider keeping margin higher.
 - Market research may help gauging customer interest in the index-linked product
- c) Marketability
 - Lack of limited pay or single pay option in the product
 - No whole-life option
 - Customer may not like to carry investment risk inherent in the index-linked product and may prefer guaranteed returns
 - The formula linking index and benefits should be simple, clear and easy to understand for the customers.
- d) Suitability to meet customer needs for a defined target market
 - Companies current customers mainly buy participating products and may prefer the relative simplicity of the new index-linked product
- e) Competitiveness
 - Lack of prior experience may lead to higher margin which can make the index-linked product uncompetitive
- f) Distribution method and remuneration
 - Existing channels (tied agents, banks and brokers) may not be used to the new proposed index-linked product
 - Additional training requirements
 - Commission and rewards need to be in line with other savings products
- g) Financing / capital requirements
 - No limited pay option which could lead to reduced NB strain, if any
 - Reinsurance may help reduce capital requirements
- h) Onerousness of options and guarantees
 - In index-linked products, guarantees accumulate over time.

- Need to test the onerousness of guarantees at different interest rates
- i) Sensitivity of profits
 - How sensitive the profit is to each economic and demographic experience. Particularly important for a new category of product.
 - Should be less sensitive to interest rate movement in comparison to non-par savings products
- j) Extent of cross subsidies
 - There may be cross-subsidies between sum-assured levels, age-wise, gender-wise, smoker/non-smoker, etc.
- k) Administration systems
 - Survival additions in the index-linked product are similar to bonuses in participating products so company may not have problem in building features in the admin system
 - Need to ensure accuracy in gathering external index data and resultant calculation of additions and their accrual with individual policies.
- l) Service standards
 - There may be queries from customers during the duration of the policy of the new index-linked product.
 - Level of complaints depend on customer's understanding of the product at point of sale.
- m) Company reputation
 - The new product should not create confusion in the minds of customers and distributors.
- n) Treating customers fairly
 - Surrender values in the index-linked may not be high enough and need to be communicated clearly.
- o) Level of risk
 - The benefits are going to be linked to an external index – need to ensure that investment of assets matches with the profile of liabilities.
 - Assumptions regarding policy discontinuance should be appropriate.
 - Need to consider how the risks from index-linked product will affect the overall risks for the company
- p) Underwriting philosophy
 - Should not be a big issue as it is expected to be in line with the existing participating products.
- q) Reinsurance terms and other benefits
 - Reinsurance support may offer some certainty for new product category. It should be easy to get the support as the mortality risk is similar to other savings products.
 - May consider financial reinsurance to reduce NB strain especially if capital is limited.
- r) Taxation
 - The product design should allow customers tax-benefits as cover is 10 times the annualized premium.

s) Legal and regulatory constraints

- Regulatory guidelines relevant to 2019 non-linked product regulations and subsequent to that must be followed.

[Note: Maximum 1.5 marks within each standard design factors; Max 6 marks for just listing all the factors]

[24]

v)

- Market Risk:
 - UL (without guarantee): no guarantee accumulation.
 Index-linked: Maturity sum-assured is guaranteed and guarantee accumulates further with increasing duration assuming survival additions cannot be negative.

[1]

- UL:
 - Fund value can go up and down and negative returns are possible. Assets and liability match perfectly by design.
 - Lower fund value would have second-order impact on the company in form of reduced fund management charges.

- Index-linked:
 - Survival additions are based on a formula and may include averaging of returns.
 - Assets and liability mismatch possible and ALM analysis and tracking become important

[1]

- Insurance Risk (Mortality, Persistency, Expense):
 - Some charges are not fully guaranteed in unit-linked products and can be revised in the future however overall cap on charges apply due to reduction-in-yield requirements.
 - Surrender charges are capped in unit-linked products in initial years and are zero after a few policy years.

[0.5]

[0.5]

[3]

vi)

Impact of index-linked product on the par products:

- Lower NB in par
 - The potential participating customers may be moved to the new product leading to reduced volume of new business from participating category
- Dwindling par book
 - Lower NB along with lapse/surrender of existing book may lead to reduced par fund or lower than expected growth of par fund.
 - More outflow than inflow can lead to liquidity challenges.
- High acquisition and maintenance expenses
 - NB volume may reduce but some of the fixed expenses may remain at the same level e.g. admin system, bonus system
- High surrenders, lapse and re-entry
 - The new product may not have just impact on NB of par products. It may also encourage existing par customers to lapse their par policies and buy

[Max 4]

index-linked policies either due to new product being more customer-friendly or distributor incentives being higher.

- Selective withdrawal of good lives may also lead to worsening mortality experience in par book

Impact of reducing volume of par book

- More volatility
 - Reduction in par fund may affect company's capacity to absorb volatility in actual experience (e.g. investment, mortality)
- Reduction in FFA, impact on solvency (ASM)
 - Indian regulations permit par FFA to be used as part of ASM which can support par and other non-par business.

[Max 2]

[6]

vii)

- Par has discretionary portion but for non-participating index-linked product:
 - Objective definition/formula how returns would link with the external index
 - Survival Additions consistent with Benefit Illustration
 - Ensuring no-error in determining the return every year
 - Appropriate systems
 - Smooth and timely servicing and claims settlement
 - Overall financial stability of the company to meet its obligations

[0.5 mark for each valid point; total maximum 2]

[2]

[50 Marks]

Solution 2:

i)

- The company has had a good market share basis its competitive price offering. Because Term Insurance is generally a vanilla offering competitive pricing would continue to be the most important factor
- Company would need to carefully choose its target segment and where to offer competitive pricing e.g.
 - Young ages Vs old ages
 - High SA or Low SA
 - Shorter Terms Vs Longer Terms
 - Regular Premium terms Vs Limited Pay Terms
- The choices would depend on company's existing customer segments that is in turn linked to distribution channels the company has. E.g.
 - Online channels and aggregators display young Male 1 Cr SA rates to attract customers for regular premium
 - Bank channels prefer Limited Pay policies
- Almost entire industry would be forced to reprice the product almost simultaneously. So company needs to decide whether to reprice immediately or wait for key competition to 'reveal' their prices. This again is dependent on whether company is competing with other insurers in a bank in open

architecture or on an online platform. Of course if the company decides to wait then it needs to weigh in lower or negative margins till that time

- Beyond Pricing there are other considerations that company may want to evaluate is
 - Company's Brand strength so that it may choose to charge a premium Vs its competition
 - Product Differentiations through any of additional options
 - Like riders
 - Income Options
 - Service Offerings like Health check ups, gym memberships
 - Additional benefits/Premium reduction through wearables like fitness bands etc
 - Service Parameters - the most important being the Claims Paid Ratio. i.e. How many claims the company pays Vs the claims it receives
 - Ease of onboarding – Essentially the documentation and UW the company requires to on board a customer. A simplified customer journey with surrogates would enable better customer acquisition. Of course there would be an interplay between UW and prices. Smarter UW may enable efficient UW without an impact on mortality. Lesser UW would have an impact on prices
 - Speed of onboarding

[0.5 mark for each valid point, Max 6 Marks]

[6]

ii)

- a)** Risks in existing protection portfolio would primarily arise from a mortality and lapse perspective
- Because the mortality experience has gone up across the industry necessitating a reinsurer price revision the company must conduct a deeper analysis of its own portfolio to assess the pockets of emerging poor experience. It is possible that segments where the industry may have been impacted could be higher or lower in the company's in-force book. Analysis below would help to ascertain the extent
 - E.g. Policy Duration wise, Channel wise, UW level wise, SA wise, Customer profile – Education/Income/Occupation
 - It is possible that portfolio of the company may be young and yet to develop completely. In such a scenario it is important to discuss with partner reinsurers in terms of what are the pockets of poor experience they are seeing for rest of the industry and accordingly map those to your own business in-force
 - Because Premium rates have gone up across the industry it is likely that customers who have locked into the current prices may not lapse their policies as expected. Generally Term insurance policies are long term lapse supported so lower than expected long term lapses would tend to lead to strain on the company.
 - Risk is compounded with both higher persistency and higher mortality.
 - Reduced value of in-force book. Potential increase in reserves and required capital.

- Other items to consider from a reinsurance perspective
Possibility of reinsurer default – Reinsurer ratings
Magnitude of reinsurer default (LGD) - Expected PV of difference between expected experience and older reinsurance rates
Lesser operating reinsurance support in terms of Claims authority, Claims approval etc
- Reinstatement Risk
Customers who have lapsed their policies may opt for revivals and revival rate may go up leading to further strain

Company would need to carefully design revival process so that unhealthy lives do not find their way back into the portfolio. A mere declaration of good health as in savings portfolio may not suffice and more extensive guidelines would need to get in place

[0.5 mark for each valid point unless stated, Max 5 Marks]

[5]

b) Risks in new protection portfolio would primarily arise from a mortality and sales perspective

- New pricing and its effect on Sales. If prices have gone up general demand may suffer as well as demand Vs competition. [0.5]
 - Lower Sales would impact Margins as well as Fixed Expense spread [0.5]
 - Average ticket size may also come down due to higher premiums. That may lead to larger proportion of retained risk. [0.5]
 - If price increase is delayed waiting for competitor pricing, company may have to absorb increased reinsurance premiums leading to reduced profitability. [0.5]
 - Whether new reinsurance rates have allowed for all emerging experience and whether the rates have become too pricey or haven't still allowed for all emerging experience [1]
- [2]**

iii)

Pros and Cons of 1 year term policy Vs Long Term policy

- 1 year term policy sold to a customer at a particular age is likely to be much less expensive Vs long term policy sold at same age because 1 year rates would not need to price future age related increases in mortality. However, if the one-year cover is more expensive due to higher initial expenses, people looking for short-term cover may buy long-term policy and lapse afterwards. [1]
- Not having high initial expenses can make product innovative and optically cheaper in the immediate however as age goes up the prices would vary with age at each renewal.
- This aspect needs to be disclosed upfront and clearly communicated because customers generally expect their policies to have same premium throughout
- 1 year product can come on top of lesser UW and therefore quick onboarding. It can be used to further upsell the customer another product [1]

- However, customer in a term product may invariably look for guaranteed renewability with no further evidence of insurability. Because as age goes up one may develop a health condition and no customer would want to be uninsured at that age. In this scenario initial UW would need to be stronger
- Needs to be decided whether same rates would be charged for new business Vs renewing business at same age
- Renewing business may have deteriorated in experience because of expiry of selection period. In such a case charging same premiums for new business as well may lead to less competitiveness
- Charging different rates may mean good lives exit and re-enter at new rates and old live experience deteriorating
- Other Points
 - Lapse risk may be lower but still there to the extent of expense recovery and selective withdrawals
 - System complexity may go up in case company wants to offer different rates between new business and different eras of renewing customer
 - Distribution compensation. Initial premium lower so lower compensation
 - Reserve built up would be lower and therefore lower investment risk
 - Conflict between current long term product and this variant. May lead to loss of volumes and margins

[0.5 Marks per bullet unless otherwise stated, Max 7 Marks]

[7]

iv) The company has a presence across all segments. Most impacted segments would be the ones heavy on protection – mortality and morbidity and therefore Individual Term, Individual Health and Group Protection. Key points to consider would be

- Whether company has a pandemic reinsurance cover or not. If yes then to what extent of losses it covers the company
- From Actual Claims intimated perspective company would need to look out for patterns such as any adverse or sudden spike in claims trend and its classification by Age/Morbidity
- Claims intimation and IBNR
 - i. Because of lockdown people may not report claims on time and their might be a surge in claims later
 - ii. For Morbidity portfolio people may postpone important but not urgent surgeries that may reduce claims in the short term but only to bounce back later
- In initial stages when there is not enough country and own data the Company would need to do extensive data analysis of its in-force portfolio in terms of
 - Age mix of protection heavy portfolio
 - Comorbidities
 - i. Can be identified by any extra loadings in data for relatively younger business
 - ii. For older business when the business is out of its UW select period this data may be less relevant

[Max 1]

[Max 2]

- Density of population that can be measured by Cities and Pin Code
- A combination of above can point to most high risk populations

Other points to consider

- Because of lockdown Travel may come down leading to lower Accidental Claims
- Economic uncertainty and job losses may lead to income and wealth erosion leading to high stress factors. This may affect physical and mental health leading to higher incidence of suicides. This would affect the company more so if policies are outside their 1 year suicide exclusion period
- Once there is vaccine the immediate impact of the pandemic may get taken care of. But with more data and further research it needs to be seen what impact long term mortality trends would see because this pandemic may alter health parameters of individuals more permanently and varying by Age

[0.5 Marks per valid point, unless specified]

[6]

v)

The company has a presence across all segments.

a) Most important risks to consider according to product LOBs apart from mortality/morbidity would be

Non Par Business

(1 Mark for each valid point, Max 4 Marks)

- Interest rate guarantees and reinvestment risk - Need to consider to what extent hedged and the residual risks in terms of flat interest rate movements (more than usual) and yield curve shape changes
- In case of derivatives same as above allowing for mitigating effects of derivatives
- Need to consider whether new derivatives would continue to be made available by counterparties in face of volatility in interest rates
- If interest rates go down what is the effect of people staying on longer on the books i.e. lower lapses because they would value their guarantees even more. In case products are lapse supported the impact would be negative
- For New business NPar business may go up or down. Whilst there would be general reduction in savings of customers but customers may want to lock in yields for NPar products leading to higher demand. Company Appetite would need to get considered

Protection Business

[Max 1.5 Marks]

- Apart from mortality risks mentioned in earlier question and interest risks above, there might be again a higher demand of protection business as more awareness dawns on people. This may lead to higher demand in
 1. Individual Term Business
 2. Individual Health Business
 3. Group Employer employee business as companies offer more term and COVID health protection to their employees

- Reinsurance support may not be easily available due to uncertainties or may be available at higher price or with more restrictive underwriting. It may have an impact on capital requirements also.
- For NB, it may be difficult to conduct medicals leading to lower volumes or higher non-medical limits.

ULIP Business

[Max 2.5 Marks]

- Although liabilities are matched with assets there would be a significant impact on fund charge earnings if equities markets stay low
- Because of lower savings and aversion to risk persistency may go down further impacting charges on this product line
- Poor PH returns may lead to customer complaints in particular if they hadn't understood the product riskiness in first place
- Because of volatility in markets customers may wish to stay away from the category itself leading to lower new volumes

Par Business

[Max 2.0 Marks]

- Although PHs share in experience it needs to be evaluated against PRE
- Different scenarios of interest rates/equities/property would have an impact on PH bonuses and that should be considered Vs PRE
- This is once in a life time event and company should consider its action Vs PRE it may create
- Par business also has a guaranteed element and that needs to be evaluated as well

Company overall

[Max 5.0 Marks]

- Because of drop in aggregate demand & uncertainty, costs may come down in short term – lower inflation, lower rents etc. But with economic stimulus the inflation may go up again
- Because of job losses people may not be able to pay their premiums. Adverse persistency may impact different LOBs differently according to the policy duration and surrender value scales
- Poor persistency may impact maintenance expense coverage for the company as per policy renewal expenses may go up
- Medical underwriting may be difficult to conduct due to lockdown impacting New Business volume
- New Business because of economic uncertainty and job losses may go down or change in its mix
- Lower New business may lead to higher overruns and losses. Costs need to be looked at in such a scenario
- Business Mix may shift from Par/ULIP to NPar/Term as customers look for more guarantees. This shift would need to get re-evaluated against company's risk appetite
- Liquidity risk due to increase in claims. Settlement of recoverable amounts from reinsurers may happen only at a set frequency.

- As interest rates go down/up company valuation would get impacted in particular for NPar and SH fund.
- Further risks due to equity in SH funds
- Corporate bonds may become riskier as slowing economy may lead to defaults and lower debt ratings

[15]

b) Stakeholder to be considered

- Regulator – IRDAI
- Shareholders and the Board
- Existing Policyholders
- New Policyholders
- Distributors
- Employees
- Reinsurers
- Third Party vendors

[1/4 Marks for each item, Max 2 Marks for listing]

Regulator – IRDAI

- Most important would be to assess solvency position of the company under various stresses on a short to medium term basis and give confidence to the IRDAI on capital position. Scenarios to be considered
 - Lower New Business
 - Higher Claims
 - Poor Persistency for a short term but perhaps better persistency for longer duration for guaranteed products
 - Products dominated by market risks of
 - Interest rates
 - Equity
 - Property
 - Corporate Bonds
 - Combination of above
- In case Capital is required need to consider sources of capital – Infusion, FinRe, Sub debt etc
- Adequacy of reserves to meet all policyholders liabilities
- Continued Operations of the Company and confidence of day to day customer redressals
 - Claim servicing
 - Query and complaint handling
 - Data Back up and disaster recovery

[0.5 Marks, Max 2.5 Marks]

Existing PHs

- Reiteration of Capital position
- Communication regarding continued operations

- Any change in operations like delays, dispatch problems etc to be highlighted
- ULIP PH communication on returns and risks and portfolio composition
- Par PH communication on bonuses
- Term Business claim paid ratio and reiteration of claim payment
- Urging persistency by reiterating solvency and need for product while highlighting risks

[0.5 Marks, Max 2.5 Marks]

SHs and Board

- Capital position and continued operations
- Impact on other profitability parameters of Value like EV, Margins and Profits
- Strategic discussions – If capital constraint then change in Mix and Volumes
- If capital rich then more market share at the cost of competition

Distributors

- Confidence to distributors on capital and continued operations
- Enabling digital and non face to face modes of selling and collections (new or existing business)
- Reiteration of safety standards in view of pandemic

Employees

- Employee safety
- Employee's concern on job security
- Employee engagement
- Enabling Work from home or work from anywhere facility

Reinsurers

- More regular sharing of data on emerging COVID claims to assess the situation
- Reinsurers would have experience across countries as well as accumulated data across Indian industry
- Reinsurer opinion on new business onboarding and extra precautions to avoid COVID risks

Vendors

- Companies have lot of dependency on third party vendors for servicing and operations
- Vendor continuity is an important aspect in terms of ability to continue operations from home
- Data security

(Max 1 Mark each for each stakeholder unless specifically stated)

[9]

[50 Marks]
